Evaluating AI Meeting Summaries with a Reusable Cross-Domain Pipeline

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Abstract

We present a reusable evaluation pipeline for generative-AI applications, instantiated here for AI meeting summaries in the Dataset Pipeline repository. The system separates reusable orchestration from task-specific semantics through five stages: source intake, structured reference construction, candidate generation, structured scoring, and persisted reporting. Unlike standalone claim scorers, it treats both ground-truth construction and evaluator output as typed, persisted artifacts that support aggregation, issue analysis, and significance testing. We benchmark the offline loop on a merged typed benchmark of 114 meetings across city_council, private_data, and whitehouse_press_briefings, yielding 340 completed meeting-model pairs and 680 judge runs for three candidate models: gpt-4.1-mini, gpt-5-mini, and gpt-5.1 (hereafter gpt-41-mini, gpt-5-mini, and gpt-51). gpt-41-mini attains the highest mean accuracy (0.583), while gpt-51 leads on completeness (0.886) and coverage (0.942). Exact paired sign tests with Holm correction show no statistically significant accuracy winner but significant retention advantages for gpt-51. A typed DeepEval baseline preserves the same retention ordering but assigns substantially higher holistic accuracy, showing why structured claim-grounded scoring is better suited to release gating and failure diagnosis. Typed analysis further isolates whitehouse_press_briefings as an accuracy-hard regime caused by unsupported specifics under a coarser reference representation. We also report a focused deployment follow-up over private_data and city_council, where the same evaluation system is reused to answer a later model-selection question and where gpt-5.4 is descriptively stronger than gpt-4.1 on all three metrics, with statistically robust gains on completeness and coverage. The offline loop is fully benchmarked here; the online loop is implemented in the deployed architecture—comprising feedback ingestion, evaluation triggering, and result persistence—but is not quantitatively evaluated in this paper.

1. Introduction

AI meeting summaries are often evaluated by prompt iteration and qualitative spot checks. That workflow is rarely sufficient for deployment. Model upgrades can silently degrade factual quality, improvements in one aspect of summary behavior can mask regressions in another, and production-facing failures are usually structured rather than anecdotal: unsupported additions, omitted actions, role confusion, or over-committed certainty.

The broader systems problem is not limited to meeting summarization. Teams typically need evaluation evidence for multiple generative-AI products, including search, summarization, assistants, and question answering. If each product builds its own evaluation stack from scratch, the organization loses comparability, inspectability, and operational reuse. This paper argues that the reusable unit should be the evaluation pipeline: artifact construction, controlled candidate generation, automated comparison, persisted explanations, and release-facing reporting. The roadmap perspective makes this even more concrete: a dependable AI quality loop needs reusable control points for data simulation or collection, dataset construction, offline benchmarking, CI/CD gating, and online feedback.

Existing evaluation frameworks often treat evaluation as a scoring step rather than as a persistent quality-control substrate, which means failures are diagnosed after the fact rather than surfaced systematically through retained artifacts, typed reports, and release-facing comparison outputs.

This work directly extends *Evaluating Embedding Models and Pipeline Optimization for AI Search Quality* by Philip Zhong, Kent Chen, and Don Wang (arXiv:2511.22240). That earlier paper established a reusable pipeline pattern for AI search through curated evaluation data, automated benchmarking, and deployment-oriented comparison of system variants. The present paper instantiates the same pattern for AI meeting summaries, where the reference artifact becomes structured meeting-summary ground truth and the comparison layer becomes claim-grounded factual scoring over generated summaries.

This continuity is repository-level rather than merely conceptual. Across the search and meeting-summary papers, the system invariants are the same: persisted artifacts, controlled batch execution, report-backed comparison, judge/result aggregation, significance reporting, and release-facing evaluation outputs. What changes across domains is the semantic layer rather than the control plane: search uses query-document relevance references and retrieval metrics, whereas meeting summarization swaps in structured GT generation, application-facing summary candidates, and claim-grounded factual evaluators. We therefore present the meeting-summary benchmark not as an unrelated application study, but as a second domain instantiation of the same evaluation substrate under a changed task-specific reference and scoring layer.

The present paper is written as a system paper. Its contribution is not a new factuality metric in isolation, but a reusable evaluation substrate implemented in the Dataset Pipeline repository. The empirical section uses the meeting-summary instantiation of that substrate to show what the pipeline produces in practice: durable artifacts, typed benchmark partitions, structured GT and structured score generation, statistically grounded release evidence, reusable error analysis, and explicit quality-loop support for benchmarking, regression detection, and failure-case discovery.

Relative to the earlier AI-search paper, the present work contributes three new system elements. First, it elevates structured GT generation into an explicit pipeline stage rather than assuming static reference data. Second, it introduces typed benchmark partitioning, which makes it possible to surface domain-specific failure regimes rather than only merged averages. Third, it adds claim-level structured scoring with formal metric definitions, which in turn enables the significance-backed model-selection analysis

in Section 6.3 and the error-structure diagnosis in Section 6.6. These additions are not cosmetic extensions of the earlier pipeline; they are the mechanisms that make the meeting-summary benchmark diagnostically useful.

The main contributions are:

1. a five-stage reusable evaluation control loop (source intake → structured reference construction → candidate generation → structured scoring → persisted reporting) that enables cross-domain reuse by separating orchestration from task-specific reference schemas, demonstrated through the meeting-summary instantiation in this paper and the companion AI-search instantiation
2. an implemented meeting-summary evaluation stack with transcript-backed storage, structured GT construction, structured score generation, candidate generation, offline evaluation, and persisted reporting
3. an expanded typed benchmark over 114 meetings and three candidate models, including dataset-type breakdowns, meeting-level significance testing, and evaluator disagreement analysis
4. a concrete pipeline-reuse demonstration showing that the same evaluation system can be reapplied without reconfiguration of judges, metrics, or evaluation protocol to a subsequent deployment question, yielding statistically robust model-selection evidence on enterprise and civic-proceedings data

2. Design Goals and Cross-Domain Reuse

The design is driven by four system goals.

2.1 G1: Inspectable Artifacts

Every major object in the pipeline should be auditable after execution: source transcript, GT, candidate summary, judge outputs, and report. This favors file-backed persistence over ephemeral evaluation runs.

2.2 G2: Reusable Orchestration, Task-Specific References

The orchestration layer should be reusable across AI tasks even when the reference artifact changes. In AI search, the reference is query-document relevance. In meeting summarization, the reference is structured GT over meeting topics, points, and decisions. The pipeline should keep the orchestration stable while swapping the task-specific reference layer and metric definitions.

2.3 G3: Release-Oriented Comparison

The pipeline should support repeated comparison of candidate systems under a controlled protocol, so that model-selection decisions are backed by persisted evidence rather than one-off manual review.

2.4 G4: Extensibility from Offline to Online Quality Loops

The quantitatively validated core in this paper is the offline loop. The current architecture also includes an implemented online path, comprising feedback ingestion, evaluation triggering, result persistence, and downstream metrics emission, in which production issues can be clustered, curated, and turned into new benchmark items, even though that path is not quantitatively characterized in the present study. In this sense, the system is designed not only for evaluation but for evaluation maintenance.

These goals yield a clean cross-domain abstraction. The reusable skeleton is:

- persist source artifacts rather than treating them as transient inputs
- construct task-specific reference artifacts under version control
- generate or ingest candidate outputs under controlled configurations
- evaluate candidates with structured, inspectable outputs
- persist reports for model selection, regression tracking, and future dataset improvement

Two properties are especially important in the present meeting-summary instantiation.

- structured GT generation: the repository does not treat GT as an opaque annotation blob; it generates and persists typed meeting objects
- structured score generation: the evaluator does not emit only a single free-form judgment; it emits claim-level alignments, per-metric scores, explanations, and issue labels that can be aggregated later

This design also aligns with the internal roadmap concepts behind the broader AI datasets and evaluation program:

- golden seed data: high-quality reference artifacts anchor trustworthy benchmarking
- dataset synthesis and quality assurance: data generation is paired with validation rather than treated as a blind scaling step
- evaluation orchestration: repeated runs are controlled by a common execution pattern
- evaluation metrics and insights: the system must produce not only scores, but diagnosis and actionable model-selection signals

2.5 Positioning Against General-Purpose Evaluation Frameworks

General-purpose evaluation frameworks such as RAGAS [10, 11], TruLens [12], and DeepEval [13] solve adjacent problems and are valuable points of comparison. RAGAS emphasizes reusable metrics and synthetic testset generation. TruLens emphasizes feedback functions, tracing, and ground-truth agreement. DeepEval emphasizes metric libraries, local test execution, and CI/CD-oriented LLM testing. The present system overlaps with all three, but its scope is different: it treats structured GT construction, structured score generation, typed benchmark partitioning, and persisted release-facing artifacts as first-class pipeline stages rather than optional surrounding infrastructure.

The comparison in Table 0 is architectural rather than exhaustive. It is intended to clarify system scope, not to claim that other frameworks cannot be extended. Capabilities evolve quickly; the table summarizes the official literature where available and otherwise the official software documentation cited in the references as accessed on April 17, 2026. We do not claim an empirical head-to-head evaluation against these systems in the present paper; that comparison is deferred to future work. Here, Partial indicates that a framework offers a functionally adjacent capability but not the specific combination of structured generation, typed partitioning, and file-backed persistence that this system treats as first-class. These ratings are based on the official literature where available and otherwise the official documentation cited in References 11-13, and they may not reflect later framework updates.

The goal of Table 0 is therefore not to claim score superiority over general-purpose evaluation frameworks. Its purpose is to clarify a systems boundary: this work treats structured reference construction, persisted artifacts, typed partitioning, and release-facing comparison as first-class infrastructure rather than as optional wrappers around a scoring step.

Capability dimension	This system	RAGAS	TruLens	DeepEval	Standalone claim scorer
File-backed end-to-end artifacts	Yes	No	Partial	Partial	No
Reusable cross-domain architecture	Yes	Partial	Partial	Partial	No
Structured GT generation stage	Yes	Partial	No	No	No
Meeting-summary-specific typed schema and metrics	Yes	No	No	Partial	Partial
Typed dataset partitions	Yes	No	No	No	No
Release-gating / regression reports	Yes	No	No	Partial	No
Multi-model comparison reports	Yes	Partial	Partial	Partial	No
CI/CD quality-gate path	Yes	No	No	Partial	No

3. System Architecture

The repository is not itself a meeting assistant product. It is the evaluation and artifact layer that supports one.

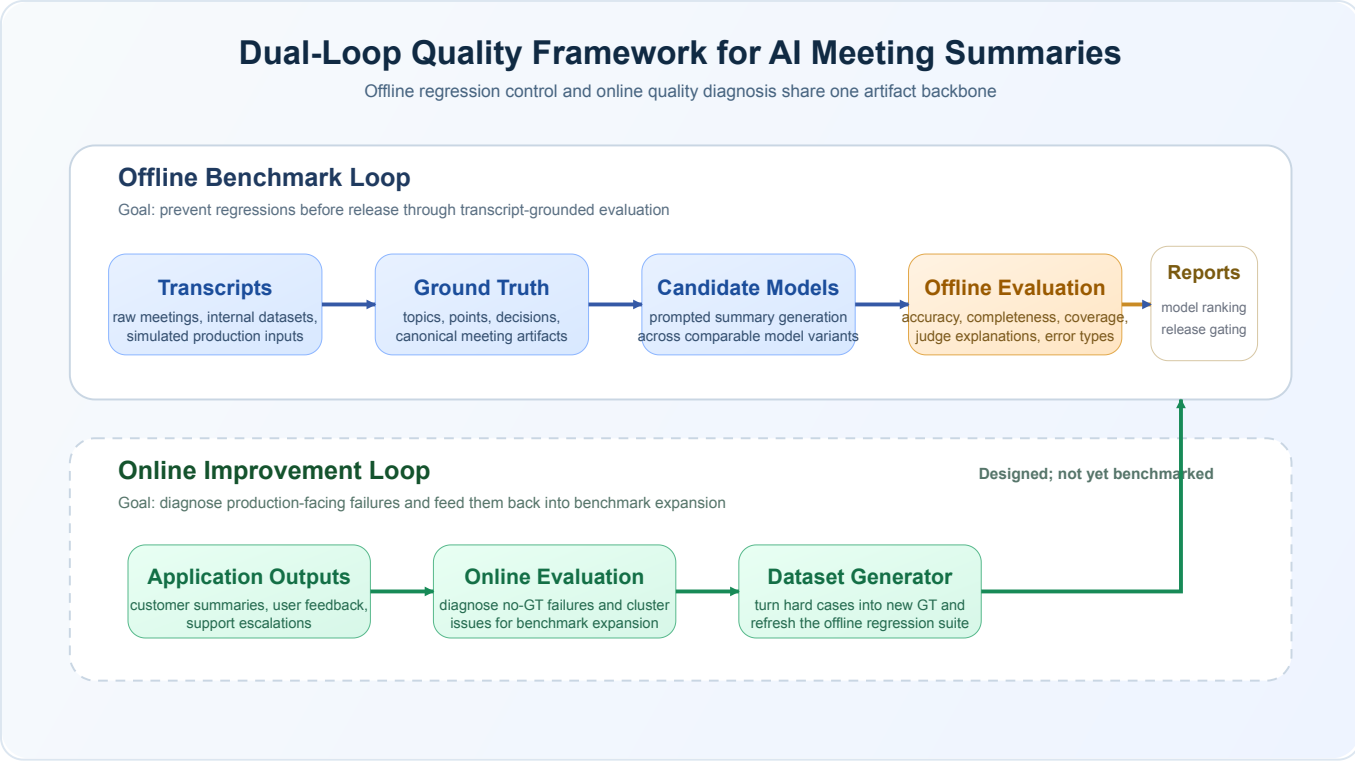

Figure 1. Reusable quality-loop architecture for AI meeting-summary evaluation. Solid components correspond to the benchmarked offline loop. The dashed online loop is implemented as system context but is not benchmarked in this paper.

Figure 1 shows the high-level architecture. Read through the roadmap lens, the loop consists of five reusable stages: a transcript simulator or production-data intake layer, a dataset generator, an offline evaluation stage, a CI/CD quality gate, and an online evaluation stage. The offline loop provides benchmark-backed regression control and is the portion validated directly in this paper. The online loop is implemented in the current system as an integrated feedback-to-evaluation path, but it is not quantitatively benchmarked in this paper. We therefore include it as deployed system context while restricting the paper's measured claims to the offline loop.

At the repository level, this translates into transcript-backed artifact storage, prompt-versioned GT construction, prompt-versioned candidate generation, offline evaluation with structured judge outputs, entry points for online and grounded evaluation, and file-backed reports that remain inspectable after execution. Taken together, these components support the three concrete benefits emphasized in the roadmap: benchmarking under a stable protocol, regression detection when models or prompts change, and failure-case discovery through persisted explanations and issue labels.

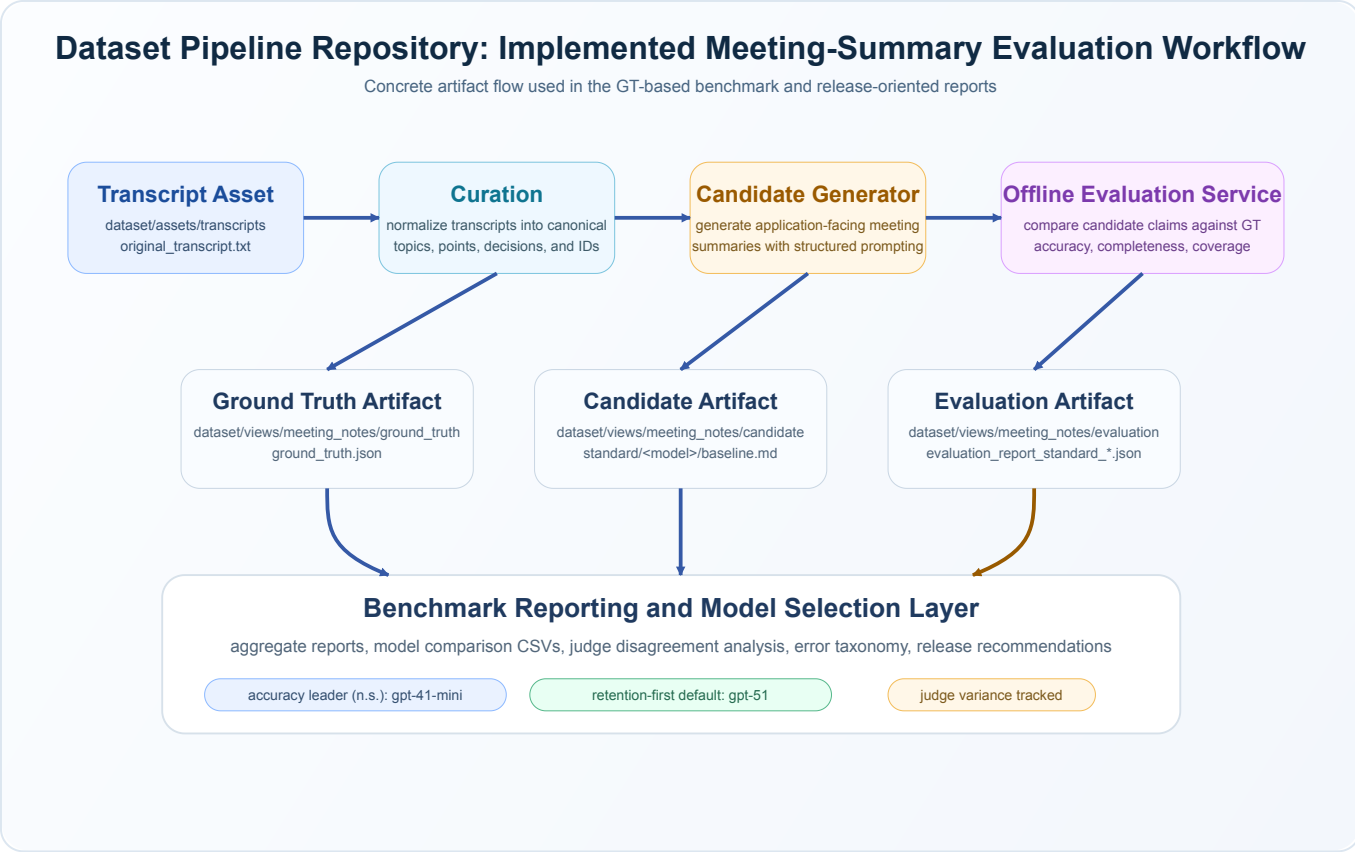

Figure 2. Repository-level realization of the five-stage control loop. Transcript assets feed both GT construction and candidate generation; offline evaluation consumes (GT, candidate) pairs and emits structured reports for comparison and release-facing analysis.

Figure 2 grounds the design in the current repository. Unlike Figure 1, which is architectural, this view maps directly to persisted artifacts and the scripts used in the benchmark workflow.

The two figures should be read together. In Figure 1, Stage 1 source intake corresponds to the transcript assets, Stage 2 structured reference construction corresponds to GT generation, Stage 3 candidate generation corresponds to application-facing summary generation, Stage 4 structured scoring corresponds to offline evaluation, and Stage 5 persisted reporting corresponds to the benchmark and release-report layer. In Figure 2, the transcript curation node takes raw transcript files as input and outputs normalized transcript assets. The GT construction node takes those transcript assets and outputs typed JSON structured around meeting_context, participants, topics, points, and decisions. The candidate-generation node takes the same transcript assets and outputs application-mirroring markdown summaries. The offline-evaluation node takes (GT artifact, candidate artifact) pairs and outputs structured evaluation JSON containing extracted claims, per-metric scores, issue labels, and explanations. This topology is deliberate: transcripts feed both GT and candidates so that reference construction and candidate generation remain comparable, while reporting stays downstream of structured scoring so that the same aggregation logic can be reused across domains.

4. Pipeline Components

4.1 Artifact Model

The repository persists meeting-summary artifacts in a file-backed layout:

```
dataset/
├── assets/transcripts/<meeting_type>/<meeting_id>/original_transcript.txt
└── views/meeting_notes/
    ├── ground_truth/<meeting_type>/<meeting_id>/meetingsummary/ground_truth.json
    ├── candidate/<meeting_type>/<meeting_id>/<variant>/<model>/baseline.md
    └── evaluation/<meeting_type>/<meeting_id>/offline/evaluation_report_<variant>.json
```

This layout matters systemically because it makes every stage independently inspectable. Transcript, GT, candidate, and evaluator outputs can all be audited or regenerated without ambiguity. It also makes cross-domain reuse practical: the repository can keep the same orchestration while swapping task-specific reference artifacts. In roadmap terms, this is what allows the same quality loop to be aligned by different teams while preserving a common evaluation substrate.

File-backed persistence is also a deliberate operational choice rather than a convenience format. Persisted files support independent inspection, partial regeneration, and version-controlled diff without requiring a live database connection months after execution. That property matters for release-facing evaluation pipelines because historical runs often need to be re-audited after the original execution context has disappeared.

The stage boundaries are similarly stable by design. The reference-construction stage consumes normalized transcript text and emits typed GT JSON; candidate generation consumes the same transcript asset and emits application-facing markdown; structured scoring consumes (GT JSON, candidate markdown) pairs and emits evaluation JSON containing metric scores, explanations, and issue labels. Prompt internals, model choices, and subset-specific heuristics are therefore implementation details behind persisted interface contracts rather than part of the orchestration boundary itself.

In this paper, candidate has a specific operational meaning: it is the application-generated meeting summary being evaluated against GT. In the current benchmark, candidates are produced by a repository pipeline that mirrors product behavior rather than by replaying production logs directly, but analytically they still represent the application's summary output.

4.2 Ground-Truth Construction

The meeting-summary GT is organized around:

- meeting_context
- participants
- topics
- points

- decisions

Point identifiers such as t_001_p_001 are assigned after the semantic structure is generated. This preserves a stable downstream schema while keeping the generation stage focused on meeting content rather than identifier formatting.

At a high level, the GT path in this paper is a two-tier quality process: automated structured extraction produces the initial reference artifact, and bounded human review or automated audit strengthens that artifact when stronger evaluation trust is required. This framing matters because downstream model scores are only as reliable as the reference they are scored against.

This is a core systems property of the paper: GT is generated as a structured representation, not merely as untyped prose. That structured representation is what later enables stable claim alignment, metric aggregation, and typed benchmark analysis.

The default GT path in the current repository is automated. GroundTruthService calls MeetingSummaryGroundTruthGenerator.generate_ground_truth_advanced(), which executes a multi-stage pipeline rather than a single-pass extraction. Stage 1 generates two independent draft GTs, one using gpt-5-2025-08-07 and one using anthropic.claude-sonnet-4-20250514-v1:0. Stage 2 aligns these drafts. Stage 3 re-reviews both uncertain and single-aligned items. Stage 4 merges the aligned and reviewed material, after which the repository assigns stable t_###_p_### and d_### identifiers and recomputes metadata counters. The exact prompt assets used for this benchmarked path are listed in Section 10.

The dual-model drafting step is a systems decision, not just a prompt flourish. Single-model GT generation risks systematic omission of items that one model is less likely to extract, and those omissions would become silent reference loss. Dual-model drafting with explicit alignment turns disagreement into a first-class quality signal that can be reviewed, merged, or audited before it propagates into downstream scoring.

All stages are executed through JSON-schema-constrained prompts and cached as intermediate artifacts. Quality control in the default path is therefore primarily machine-based: schema-constrained generation, dual-model draft comparison, targeted review of uncertain items, identifier normalization, metadata recomputation, and an optional automated-audit path all act as guards on GT consistency. The repository also exposes an optional audit stage that scores GT faithfulness and coverage against the transcript, but that audit is not a mandatory gate for every item in the present benchmark. One subset is stronger than this default path: all 22 private_data GT artifacts in the merged typed benchmark underwent human review after automated construction. The benchmark should therefore be interpreted as using machine-generated structured GT overall, with partial automated-audit retention across slices and an additional human-review layer for the full private_data subset.

The same automated GT generation pipeline is used across city_council, private_data, and whitehouse_press_briefings; there are no subset-specific model swaps in the reported benchmark. The main subset differences arise after generation. private_data receives an additional human-review pass over all benchmark items, whereas city_council and whitehouse_press_briefings remain on the

automated path with optional automated audit only. In that stronger path, human review should be understood as rubric hardening, not as fully manual re-authoring from scratch. Reviewers inspect the same transcript against the structured GT draft, check for missing decisions, broken identifiers or schema structure, weak context, and topic-boundary errors, and then apply targeted corrections to move the artifact toward golden GT suitable for downstream scoring. The goal is not to replace automated GT generation with manual curation everywhere, but to make automated GT accurate enough that human review remains bounded. A second difference is artifact normalization. city_council and private_data GT files persist metadata counters directly, whereas all whitehouse_press_briefings GT files retain the same structured topics and points but omit stored metadata counters, so evaluation traverses topics, points, and decisions at runtime. These distinctions matter for interpretation and are revisited in Sections 6.2 and 6.7.

Two transparency notes matter for interpretation. First, the repository now retains enough alignment artifacts to quantify GT agreement for the full city_council slice. The exported report gt_agreement_stats_all_20260421.csv shows retained stage-2 alignment artifacts for all 34 city_council meetings. Averaged per meeting, those stage-2 reports record 15.91 fully aligned items, 13.53 single-aligned items, and 3.76 items sent to explicit review, while the final merged GT averages 32.79 output items. We report these as separate stage summaries rather than as a strict additive partition of final GT items, because review and merge can combine, drop, or re-express material before the final artifact is written. The follow-up review stages are conservative rather than cosmetic: reviewed single-aligned items are almost always kept or modified (11.35 on average versus only 0.18 dropped), while reviewed uncertain items are usually merged (3.03) rather than excluded (0.59). These retained alignment reports provide a quantitative proxy for GT difficulty on the city_council slice. Equivalent stage-2 temp artifacts are not retained in the current workspace for private_data or whitehouse_press_briefings, so benchmark-wide agreement percentages cannot be reconstructed for those subsets from historical runs alone. The whitehouse_press_briefings slice is therefore the least transparent GT-protocol slice in the current benchmark: its GT artifacts are structured and evaluable, but the historical runs do not retain enough intermediate protocol traces to reconstruct agreement or audit difficulty after the fact. Second, retained audit artifacts are present for 20 meetings, but the audit stage was not enforced as a mandatory gate across all benchmark items. In other words, the benchmark is based on structured GT that is predominantly machine-generated, partially automated-audited, additionally human-reviewed for the full private_data slice, and quantitatively agreement-instrumented for the full city_council slice rather than universally audit-gated or universally human-reviewed.

4.3 Candidate Generation

For this study, each meeting is summarized under the same candidate pipeline:

- summary style: standard
- candidate filename: baseline.md
- fixed prompt family: standard_v1.yaml + format_v1.yaml

This design isolates model choice under a fixed prompt pipeline rather than conflating model differences with prompt tuning. The main merged benchmark uses three candidate-generation models, and the targeted deployment follow-up in Section 6.2.1 reuses the same prompt family with two additional production-facing aliases:

Report label	Backing model
gpt-41-mini	gpt-4.1-mini
gpt-5-mini	gpt-5-mini-2025-08-07
gpt-51	gpt-5.1
gpt-41	gpt-4.1
gpt-54	gpt-5.4

4.4 Offline Evaluator and Metrics

The offline evaluator compares candidate summaries against GT through three factual metrics. The pipeline first extracts candidate claims with a fixed claim-extraction prompt family (claims_v2.yaml) and a fixed extractor model (gpt-5-2025-08-07). Those claims are then scored against GT by metric-specific evaluators.

These three metrics are chosen to separate three failure modes that merged holistic scoring hides. Accuracy captures unsupported additions and factual drift, Coverage captures omissions of GT content, and Completeness captures how fully the covered material is retained. A single scalar score can reward a summary that covers few items deeply or one that covers many items sparsely without distinguishing those behaviors; the metric triplet keeps those failure modes observable.

Metric	Operational meaning
Accuracy	The fraction of extracted application-summary claims judged accurate when compared to GT
Completeness	For GT items that are already covered, the average proportion of each matched item's content retained in the application summary
Coverage	The fraction of GT items judged as covered by the candidate summary

These conceptual definitions are implemented through structured verdicts rather than free-form scoring. Let C = {c1, ..., cm} be the extracted claim set from a candidate summary and G = {g1, ..., gn} be the GT point/decision set used by the evaluator. Each candidate claim receives a verdict v(ci) in {accurate, inaccurate}. Each GT item receives a coverage state s(gj) in {covered, uncovered}. For covered GT items, the completeness evaluator additionally assigns a detail score d(gj) in [0, 1] and a detail level l(gj) in {rich, adequate, sparse, barebone}.

```
Accuracy(C, G)      = (1 / m) ∗ Σ 1[v(ci) = accurate]
Completeness(C, G)  = (1 / |Gcov|) ∗ Σ d(gj),   Gcov = {gj : s(gj) = covered}
Coverage(C, G)      = (1 / n) ∗ Σ 1[s(gj) = covered]
```

This means that Coverage is a binary core-content metric, whereas Completeness is a detail-retention metric conditioned on GT items that were already marked covered. When no GT items are covered, completeness is defined operationally as 0. Because completeness is conditional on coverage, the two metrics are not recall variants of one another: a model can cover many GT items but still summarize them sparsely, in which case coverage can be numerically higher than completeness. Note also that completeness and coverage are not directly comparable in magnitude. Coverage can exceed completeness when covered GT items are summarized with low detail scores, as observed for gpt-5-mini and gpt-51 in Table 1. Conversely, completeness can exceed coverage only in the degenerate sense that a smaller covered subset happens to receive relatively high average detail scores, a pattern visible for gpt-41-mini, which covers fewer GT items but describes the items it does cover with somewhat higher detail fidelity.

For interpretation, the repository supports a three-way reading:

- fully captured: the GT item is covered and receives a rich or adequate detail level
- partially captured: the GT item is covered but receives a sparse or barebone detail level
- missing: the GT item is uncovered

A simple example is a GT point such as “The committee approved the extension through June 2026.” A candidate that says “The committee approved the extension through June 2026” is fully captured; “The committee approved the extension” is partially captured because it omits the deadline; and a candidate that never mentions the extension is missing. Operationally, the first two cases are both counted as covered by the coverage evaluator, but they receive different completeness detail scores.

The framework remains compatible with claim-based evaluation: candidate summaries are decomposed into factual units and aligned against grounded reference content. The system-level contribution is that claim-grounded checking is embedded in a larger operational artifact pipeline rather than used only as a standalone scorer.

Equally important, score generation is itself structured. The evaluator does not stop at a single scalar judgment. It produces:

- metric-specific scores
- claim-alignment outputs
- explanations for each metric
- issue labels such as unsupported_by_gt, fabricated_facts, and missing

This structure is what makes later aggregation, significance testing, typed dataset comparison, and release reporting possible.

5. Evaluation Methodology

5.1 Benchmark Universe

The merged typed benchmark used in this study contains 114 unique meetings across three dataset types:

- city_council: 34 meetings
- private_data: 22 meetings
- whitehouse_press_briefings: 58 meetings

At the meeting-model-pair level, the merged report contains:

- 340 completed meeting-model pairs
- 680 evaluator runs

The count is not 342 because one city_council meeting currently contains only the gpt-41-mini candidate/report pair; the gpt-5-mini and gpt-51 outputs are missing for that meeting. As a result, the overall benchmark contains 114 meetings, but the paired three-model comparisons in Section 6.3 operate on 113 meetings.

The typed structure of the benchmark matters for system evaluation. It allows the same pipeline to be assessed not only overall, but also by dataset type, which exposes where the system is stable across categories and where task difficulty shifts. The following table summarizes the benchmark characteristics that are most relevant for later interpretation; the averages are approximate and are derived from the merged benchmark artifacts discussed in Sections 6 and 7.

Dataset type	Meetings	Avg GT items / meeting	Avg claim-to-GT ratio	Primary domain
city_council	34	~30.15	~1.02	local government proceedings
private_data	22	~36.36	~1.05	internal enterprise meetings
whitehouse_press_briefings	58	~26.34	~1.41	high-density public-affairs press briefings

The high claim-to-GT ratio in whitehouse_press_briefings (1.41 versus roughly 1.02-1.05 for the other types) is the quantitative signature of the accuracy anomaly analyzed in Section 6.2.

5.2 Judges and Aggregation

Each candidate summary is evaluated against GT by two judges:

- gpt-5-2025-08-07

- anthropic.claude-sonnet-4-20250514-v1:0

This yields:

- 340 completed meeting-model pairs
- 680 evaluator runs

For each meeting-model pair, the pipeline stores:

- judge-specific scores
- explanations for each metric
- structured issue labels
- averaged accuracy_avg, completeness_avg, and coverage_avg

We use two judges because accuracy judgments are materially noisier than a release-facing scalar should hide. As shown later in Section 6.5, mean absolute inter-judge disagreement reaches 0.106-0.141 on accuracy depending on the candidate model. A single-judge pipeline would absorb that uncertainty into the score; a two-judge pipeline keeps the disagreement visible and makes borderline accuracy calls less brittle for release decisions.

The aggregate report used in this paper is:

- meeting_notes_model_comparison_combined_20260417.csv

Artifact details for the focused gpt-5.4 versus gpt-4.1 follow-up are listed in Section 10.

5.3 Evaluation Protocol

The offline evaluation protocol follows five steps.

1. For each transcript-GT pair, the repository generates one candidate summary per model under the same prompt family.
2. The evaluator decomposes the candidate summary into factual units suitable for claim-level comparison.
3. Two judges independently align those candidate claims against GT topics, points, and decisions.
4. The pipeline stores judge-specific scores, explanations, and issue labels.
5. The benchmark report averages the two judges per meeting-model pair and then aggregates those averages across the merged typed benchmark.

5.4 Statistical Testing

To complement descriptive averages, we perform exact paired two-sided sign tests on meeting-level model comparisons for each metric. The comparison unit is the meeting. For any model pair and metric, each meeting contributes one win, one loss, or one tie; ties are excluded from the test. Because each metric induces three pairwise model comparisons, we apply Holm correction within each metric family.

This test is intentionally conservative. It does not use score magnitude; it asks whether one model wins significantly more meetings than another under the paired benchmark design. That framing fits the system use case because model releases are decided over repeated meeting-level comparisons rather than over a single pooled score.

The significance summary used in this paper is:

- meeting_notes_model_comparison_combined_significance_20260417.csv

Unless otherwise noted, descriptive benchmark means reported in Sections 6.1 through 6.6 are recomputed from meeting_model rows and rounded to three decimals. Derived delta columns are also rounded after computation rather than obtained by subtracting already rounded displayed values. We use that single aggregation rule throughout the manuscript so that the main benchmark tables and the DeepEval comparison remain numerically aligned.

5.5 Operational Telemetry

The repository is instrumented for operational-cost reporting through structured logging in log_config.metrics, which records job duration, API latency, LLM token usage, and LLM latency by stage. In addition to that instrumentation path, the repository retains four historical batch logs from which we can recover a measured subset of the evaluation-stage telemetry. We therefore report measured latency/token summaries for the stages that have explicit completion lines in those retained logs, while still acknowledging that the historical benchmark does not preserve a complete end-to-end measured cost ledger for every pipeline stage.

Table 5 summarizes the measured telemetry extracted from retained logs.

Retained stage	Samples	Avg total time (s)	Median (s)	P90 (s)	Avg total tokens
general_evaluation	49 calls	14.84	14.31	22.40	5,902
accuracy_evaluation	39 calls	47.01	37.99	86.19	10,755
completeness_evaluation	39 calls	71.03	71.24	118.78	12,384
coverage_evaluation	39 calls	65.66	54.16	128.37	11,817
offline_evaluation_job	19 jobs	476.84	531.57	607.60	n/a

general_evaluation appears in the repository and in retained telemetry because the codebase implements a broader evaluation stage family, but it is not one of the benchmark metrics reported in this paper. All headline benchmark tables and significance analyses in this manuscript use only the three factual metrics defined in Section 4.4: accuracy, completeness, and coverage.

These measured rows come from explicit evaluator-completion and offline-job completion lines in retained batch logs, not from reconstructed estimates. Within that retained slice, gpt-5-2025-08-07 is consistently slower than Claude Sonnet 4 across all four offline-evaluation stages; for example, average

accuracy_evaluation time is 62.53s for gpt-5-2025-08-07 versus 30.67s for Claude Sonnet 4, and average coverage_evaluation time is 89.53s versus 40.54s.

The measured telemetry artifact used here is:

- measured_stage_level_telemetry_from_logs_20260421_131115.csv

This table should still be read as partial rather than complete. The retained logs give high-confidence measured coverage for offline evaluation, but they do not preserve comparably clean benchmark-wide completion traces for GT construction and candidate generation. For that reason, the paper can now report measured telemetry for the retained offline-evaluation stages, but not a fully measured end-to-end cost ledger for the entire five-stage pipeline. As a rough operational reference for the missing stages, a single meeting's full offline pipeline still typically involves on the order of 31-33 LLM calls across GT construction, three-model candidate generation, claim extraction, and two-judge metric scoring. This estimate breaks down approximately as GT construction (4-6 calls for dual-model draft generation, alignment, optional review passes, and merge), candidate generation (6 calls, two prompt stages for each of the three models), claim extraction (3 calls, one per candidate), and offline evaluation (18 calls, two judges × three models × three metric evaluators). Actual counts vary with meeting length and with the number of uncertain GT items requiring Stage 3 re-review.

For the focused gpt-5.4 versus gpt-4.1 comparison added in this version of the paper, we also package a run-level operational report that records the observed rerun window, concurrency, reuse boundary, and clean completion status of the latest 2026-04-22 batch execution. In brief, that follow-up was finalized as a 2h 14m rerun at concurrency 6, reusing 21 previously completed artifacts, newly executing 91 tasks, and closing with 0 failed tasks, 0 exception tasks, and 0 missing reports:

- run_report_20260422.md

6. Experimental Evaluation

6.1 Main Benchmark Results

Table 1 reports the main merged typed benchmark result over 114 meetings, 340 completed meeting-model pairs, and 680 evaluator runs.

Model	Meetings	Evaluator runs	Accuracy	Completeness	Coverage	Mean of 3 metrics
gpt-41-mini	114	228	0.583	0.814	0.804	0.734
gpt-5-mini	113	226	0.574	0.843	0.875	0.764
gpt-51	113	226	0.553	0.886	0.942	0.794

The primary pattern is still a structured trade-off, but the larger mixed benchmark changes the frontier. gpt-41-mini now has the highest mean accuracy, while gpt-51 remains clearly best on completeness and coverage. gpt-5-mini occupies the middle ground on all three metrics: it neither leads on accuracy nor matches gpt-51 on retention.

6.2 Results by Dataset Type

Because the benchmark is typed, the merged result can also be broken down by data category.

Dataset type	Model	Meetings	Accuracy	Completeness	Coverage
city_council	gpt-41-mini	34	0.703	0.775	0.775
city_council	gpt-5-mini	33	0.709	0.820	0.860
city_council	gpt-51	33	0.688	0.878	0.896
private_data	gpt-41-mini	22	0.704	0.805	0.838
private_data	gpt-5-mini	22	0.724	0.850	0.869
private_data	gpt-51	22	0.700	0.889	0.957
whitehouse_press_briefings	gpt-41-mini	58	0.465	0.841	0.807
whitehouse_press_briefings	gpt-5-mini	58	0.440	0.854	0.886
whitehouse_press_briefings	gpt-51	58	0.420	0.889	0.963

Two patterns stand out. First, the ranking on completeness and coverage is stable across all three dataset types: gpt-51 is consistently strongest, followed by gpt-5-mini, then gpt-41-mini. Second, accuracy is much more dataset-sensitive. The whitehouse_press_briefings slice is substantially harder on accuracy for all models, which is what shifts the overall merged typed benchmark away from the earlier smaller benchmark's accuracy ordering. By contrast, the private_data slice remains comparatively accuracy-friendly even though it is fact-dense, a result that should be read together with the stronger GT protocol for that slice: all 22 private_data GT artifacts were human-reviewed after automated construction.

The systems value of typed partitioning becomes clearer when contrasted with a merged-only view:

Reporting view	Accuracy values observed	Likely release signal
Merged-only benchmark	0.553-0.583 across the three models	Moderate spread; no slice-specific alarm is visible
Typed city_council view	0.688-0.709	Acceptable accuracy regime

Reporting view	Accuracy values observed	Likely release signal
Typed private_data view	0.700-0.724	Acceptable accuracy regime
Typed whitehouse_press_briefings view	0.420-0.465	Distinct failure regime requiring targeted intervention

A release policy operating on merged averages alone would observe only a moderate accuracy spread and might not flag the White House slice as requiring targeted intervention. Typed partitioning makes that regime visible as a structurally distinct problem rather than as a statistical outlier buried inside the merged benchmark.

This anomaly is important because it is not explained by larger GT size. whitehouse_press_briefings averages about 26.34 GT items per meeting, compared with roughly 30.15 for city_council and 36.36 for private_data. The harder accuracy behavior instead comes from a denser stream of unsupported specifics: in the White House slice, the dominant issue labels are unsupported_by_gt (3930) and fabricated_facts (2531), far outweighing direct contradiction. These dataset-level counts come from the same raw claim-level issue_type aggregation over persisted accuracy_explanation payloads that later yields the benchmark-wide totals in Section 6.6. Section 7.1 analyzes the mechanism behind this regime in detail and explains why the same prompt family behaves much better on private_data and city_council.

6.2.1 System Reuse for a Deployment Follow-up: gpt-5.4 vs gpt-4.1 on Operational Slices

To separate deployment-facing model choice from the broader three-model mixed benchmark, we also ran a targeted follow-up comparison on the two operationally important non-White-House slices only: private_data and city_council. This follow-up was executed after the main merged benchmark and was not folded into Tables 1-2, because it addresses a narrower operational question and was only run on those two slices rather than on the full 114-meeting benchmark universe. Its role in this paper is therefore supplementary and system-oriented: it demonstrates operational reuse of the same evaluation pipeline under a different deployment question, using the same judges, metrics, and evaluation protocol without reconfiguration of the evaluation logic itself. The focused benchmark covers 56 meetings, 112 completed meeting-model tasks, and 224 evaluator runs under the same two-judge protocol used elsewhere in the paper.

Scope	Model	Meetings	Evaluator runs	Accuracy	Completeness	Coverage	Mean of 3 metrics
overall	gpt-4.1	56	112	0.739	0.789	0.794	0.774

Scope	Model	Meetings	Evaluator runs	Accuracy	Completeness	Coverage	Mean of 3 metrics
overall	gpt-5.4	56	112	0.754	0.844	0.868	0.822
city_council	gpt-4.1	34	68	0.745	0.781	0.781	0.769
city_council	gpt-5.4	34	68	0.760	0.831	0.834	0.808
private_data	gpt-4.1	22	44	0.729	0.801	0.816	0.782
private_data	gpt-5.4	22	44	0.745	0.865	0.920	0.843

Unlike the broader mixed benchmark, this focused comparison does not expose an accuracy-versus-retention trade-off. gpt-5.4 is descriptively better than gpt-4.1 on all three metrics overall and in both constituent dataset types. At the meeting level, the accuracy advantage remains directional rather than statistically decisive (30 wins for gpt-5.4, 24 for gpt-4.1, 2 ties; exact paired sign test $p = 0.4966$), but the retention gains are broad and robust: completeness favors gpt-5.4 in 52 meetings versus 3 (1 tie, $p = 1.54 \times 10^{-12}$), and coverage favors gpt-5.4 in 41 meetings versus 8 (7 ties, $p = 1.96 \times 10^{-6}$). Operationally, this follow-up also closed cleanly as a rerun on 2026-04-22: concurrency was raised from 2 to 6, 21 previously completed artifacts were reused, 91 tasks were newly executed, and the run finished with 0 failed tasks, 0 exception tasks, and 0 missing reports.

The private_data slice shows the largest practical gain, especially on coverage (0.920 versus 0.816, a +0.104 delta). That is operationally important because this slice contains dense enterprise coordination content and uses human-reviewed GT throughout. The improvement mechanism is also more informative than a simple “more conservative model” story. gpt-5.4 does not win by saying less. It emits more total evaluable claims than gpt-4.1, yet still achieves a slightly lower inaccurate-claim rate while covering substantially more GT content. Table 2A summarizes those mechanism-level diagnostics.

Scope	Model	Inaccurate-claim rate	Covered-point rate	Share of covered GT items judged rich
overall	gpt-4.1	0.259	0.800	0.242
overall	gpt-5.4	0.247	0.894	0.477
city_council	gpt-4.1	0.253	0.793	0.239

Scope	Model	Inaccurate-claim rate	Covered-point rate	Share of covered GT items judged rich
city_council	gpt-5.4	0.242	0.868	0.477
private_data	gpt-4.1	0.266	0.808	0.245
private_data	gpt-5.4	0.254	0.926	0.476

Paired meeting-level sign tests on these mechanism indicators point in the same direction. Overall, lower inaccurate-claim rate favors gpt-5.4 in 30 meetings versus 24 for gpt-4.1 (2 ties, $p = 0.4966$), covered-point rate favors gpt-5.4 in 41 meetings versus 8 (7 ties, $p = 1.96 \times 10^{-6}$), and rich-share favors gpt-5.4 in 50 meetings versus 4 (2 ties, $p = 3.80 \times 10^{-11}$). The same pattern holds by slice: rich-share favors gpt-5.4 in 31 versus 1 city_council meetings (2 ties, $p = 1.54 \times 10^{-8}$) and 19 versus 3 private_data meetings ($p = 8.55 \times 10^{-4}$). In other words, the gain is not merely more recall; it is higher-information retention with no directional collapse in accuracy. The dominant remaining accuracy-side failure mode in this focused slice is still unsupported_by_gt, not direct contradiction, so the comparison remains consistent with the paper's broader claim-grounded diagnosis. Artifact details for this supplementary follow-up are collected in Section 10.

6.3 Meeting-Level Wins and Statistical Significance

Aggregate means can hide whether gains are broad or driven by a few outliers. We therefore report meeting-level wins over the 113 meetings for which all three model outputs are available in the merged report.

Metric	gpt-41-mini wins	gpt-5-mini wins	gpt-51 wins	Ties
Accuracy	48	36	26	3
Completeness	3	9	100	1
Coverage	0	10	82	21

The completeness and coverage advantages of gpt-51 remain broad meeting-level effects rather than mean-value artifacts. Accuracy is different: the three models remain much closer, and the merged typed benchmark no longer supports a statistically decisive winner.

Table 2 reports the paired sign tests. Positive mean differences indicate that model A outperforms model B on the given metric. Mean-difference values are computed as per-meeting averages rather than as differences of aggregated means. Ties are excluded from the sign tests. Holm correction is applied within each metric family across the three pairwise comparisons.

Metric	Comparison	Mean diff (A-B)	A wins	Ties	B wins	Holm-adjusted p	Significant at 0.05
Accuracy	gpt-41-mini vs gpt-5-mini	0.006	60	2	51	0.447806	No
Accuracy	gpt-41-mini vs gpt-51	0.027	69	1	43	0.053297	No
Accuracy	gpt-5-mini vs gpt-51	0.021	64	2	47	0.256951	No
Completeness	gpt-41-mini vs gpt-5-mini	-0.028	20	2	91	5.25×10^{-12}	Yes
Completeness	gpt-41-mini vs gpt-51	-0.071	3	1	109	2.71×10^{-28}	Yes
Completeness	gpt-5-mini vs gpt-51	-0.043	9	1	103	4.63×10^{-21}	Yes
Coverage	gpt-41-mini vs gpt-5-mini	-0.072	15	21	77	3.26×10^{-11}	Yes
Coverage	gpt-41-mini vs gpt-51	-0.139	3	11	99	2.09×10^{-25}	Yes
Coverage	gpt-5-mini vs gpt-51	-0.067	11	17	85	5.10×10^{-15}	Yes

This result sharpens the descriptive interpretation. The apparent overall accuracy lead of gpt-41-mini is not statistically significant after Holm correction. By contrast, the completeness and coverage advantages of gpt-51 are statistically significant against both alternatives, and gpt-5-mini also significantly exceeds gpt-41-mini on those retention-oriented metrics.

6.4 Typed Empirical Baseline with DeepEval

To complement the architectural comparison in Table 0, we ran a typed empirical baseline against DeepEval on the same packaged benchmark slices. The comparison uses deepeval_typed_empirical_comparison.py and the resulting report deepeval_typed_empirical_comparison_all_20260421.csv. Unlike our system, which evaluates claim-extracted candidate summaries against structured GT with two judges (gpt-5-2025-08-07 and anthropic.claude-sonnet-4-20250514-v1:0) and three persisted metrics, DeepEval applies single-judge GEval prompts over whole summary-to-reference comparisons using gpt-4.1-mini as the evaluator model. The DeepEval prompts are also holistic rather than claim-structured by framework design: the accuracy criterion asks whether the actual summary is globally faithful to the expected GT and penalizes contradictions, unsupported additions, fabricated details, and unjustified certainty, while the coverage criterion asks how well the actual summary covers important information in the expected GT

and emphasizes actions, dates, decisions, quantities, and omissions. To keep the semantics explicit, we refer to the baseline outputs as deepeval_holistic_accuracy and deepeval_holistic_coverage; these are not metric-identical to the repository's accuracy, coverage, and completeness.

The typed comparison now covers the full merged benchmark universe: 340 completed meeting-model pairs (100 city_council, 66 private_data, and 174 whitehouse_press_briefings). The city_council slice contributes 100 rather than 102 meeting-model pairs for the same reason discussed in Section 5.1: one meeting in the merged benchmark contains only the gpt-41-mini candidate/report pair and lacks the gpt-5-mini and gpt-51 outputs. We therefore treat the DeepEval result primarily as a ranking-consistency baseline rather than as a score-calibrated head-to-head comparison. Table 3 summarizes the per-slice comparison.

Dataset type	Model	System acc.	DeepEval holistic acc.	Acc. delta	System cov.	DeepEval holistic cov.	Cov. delta
city_council	gpt-41-mini	0.703	0.878	+0.175	0.775	0.868	+0.093
city_council	gpt-5-mini	0.709	0.907	+0.198	0.860	0.916	+0.056
city_council	gpt-51	0.688	0.971	+0.284	0.896	0.965	+0.069
private_data	gpt-41-mini	0.704	0.923	+0.219	0.838	0.902	+0.064
private_data	gpt-5-mini	0.724	0.918	+0.195	0.869	0.905	+0.036
private_data	gpt-51	0.700	0.976	+0.276	0.957	0.966	+0.009
whitehouse_press_briefings	gpt-41-mini	0.465	0.911	+0.446	0.807	0.897	+0.090
whitehouse_press_briefings	gpt-5-mini	0.440	0.936	+0.496	0.886	0.937	+0.051
whitehouse_press_briefings	gpt-51	0.420	0.986	+0.565	0.963	0.982	+0.020

Three findings matter. First, the baseline preserves the same retention ordering as the repository's structured evaluation: gpt-51 is the strongest coverage-oriented model in all three slices, while gpt-41-mini is the weakest. Second, DeepEval assigns uniformly higher holistic accuracy than the repository's claim-grounded accuracy, with the largest gap appearing on whitehouse_press_briefings. Across the full benchmark, these overall deltas are computed as meeting-model-pair-weighted averages over all completed pairs rather than as simple averages of the three slice means; under that weighting, the holistic-accuracy delta ranges from +0.321 for gpt-41-mini to +0.427 for gpt-51, and in the White House slice alone it reaches +0.446 to +0.565. Third, the White House anomaly remains visible but changes character under the baseline: rather than exposing a sharp GT-grounded accuracy collapse, DeepEval treats the same richer summaries as broadly excellent. This is precisely the systems distinction the paper is making. Holistic reference scoring is useful as a lightweight sanity-check baseline, but structured claim-grounded scoring is better suited for release gating and failure-mode diagnosis because it penalizes unsupported specifics that holistic scorers tend to absorb into a globally coherent faithfulness judgment.

6.5 Judge Variance and Minimal Single-vs-Two-Judge Ablation

Because the benchmark uses two judges, model means should be interpreted together with judge spread.

Model	Judge	Accuracy	Completeness	Coverage
gpt-41-mini	Claude Sonnet 4	0.628	0.822	0.824
gpt-41-mini	GPT-5 judge	0.540	0.807	0.784
gpt-5-mini	Claude Sonnet 4	0.616	0.836	0.897
gpt-5-mini	GPT-5 judge	0.531	0.850	0.854
gpt-51	Claude Sonnet 4	0.590	0.865	0.948
gpt-51	GPT-5 judge	0.515	0.906	0.937

Claude remains systematically more generous on accuracy, while GPT-5 is somewhat more favorable to gpt-51 on completeness. The mean absolute judge disagreement per meeting-model pair is:

Model	Accuracy disagreement	Completeness disagreement	Coverage disagreement
gpt-41-mini	0.106	0.039	0.069
gpt-5-mini	0.122	0.033	0.058
gpt-51	0.141	0.044	0.018

The main systems implication is that accuracy remains the least stable metric across judges, while coverage is comparatively robust.

We can make that point more concrete through a minimal winner-only ablation built from the merged benchmark report and exported in single_vs_two_judge_ablation_all_20260421.csv. Table 4 summarizes the model that wins each dataset slice under single-judge and two-judge aggregation.

Dataset type	Claude single-judge accuracy winner	GPT-5 single-judge accuracy winner	Two-judge accuracy winner	Two-judge completeness / coverage winner
city_council	gpt-5-mini	gpt-41-mini	gpt-5-mini	gpt-51
private_data	gpt-5-mini	gpt-5-mini	gpt-5-mini	gpt-51
whitehouse_press_briefings	gpt-41-mini	gpt-41-mini	gpt-41-mini	gpt-51

The ablation reinforces two operational points. Accuracy ordering is judge-sensitive in city_council, stable in private_data, and stable but low-valued in whitehouse_press_briefings. By contrast, the retention-oriented conclusion is invariant across all three slices: gpt-51 remains the completeness and coverage leader regardless of whether the benchmark uses one judge or the two-judge average. In other words, two-judge aggregation matters most for stabilizing borderline accuracy choices, not for overturning the benchmark's stronger retention conclusions.

The city_council instability is not merely symbolic. Under Claude, gpt-5-mini outruns gpt-41-mini on accuracy (0.748 vs 0.714), whereas under GPT-5 the ordering reverses (0.693 vs 0.670). The two-judge average narrows that difference to 0.709 versus 0.703, which is exactly why this slice is the clearest example of two-judge aggregation acting as a stabilizer for borderline accuracy calls rather than as a source of fundamentally new retention conclusions.

6.6 Error Structure

Persisted evaluator outputs make it possible to aggregate failure modes across the full benchmark.

Issue type	Count across all inaccurate claims
unsupported_by_gt	5619
fabricated_facts	3224
factual_error	537
changed_certainty	131
contradicts_gt	114
changed_nature	113

The accuracy-side issue table is aggregated directly from claim-level issue_type assignments in the 678 evaluator outputs that contain persisted accuracy_explanation payloads; two White House evaluator runs record evaluation_error instead and therefore do not contribute issue labels.

For coverage, the dominant failure mode is:

Issue type	Count across all uncovered GT items
missing	2200

These counts show that the benchmark is not mostly penalizing direct contradiction. The dominant accuracy failure is adding content unsupported by GT, whereas the dominant coverage failure is omission.

The coverage-side table is aggregated from all 680 evaluator outputs with persisted coverage_explanation payloads, so the missing = 2200 total is benchmark-wide rather than filtered to the 678 accuracy-bearing outputs. The concentration of those errors is highly uneven across dataset types. The whitehouse_press_briefings slice alone contributes 3930 of the 5619 unsupported_by_gt labels, 2531 of the 3224 fabricated_facts labels, and 1106 of the 2200 missing labels, confirming that this error regime is substantially concentrated in the public-affairs domain rather than being evenly distributed across the benchmark.

The same pattern appears in model-level counts aggregated at the meeting-model level.

Model	Avg inaccurate claims / meeting-model pair	Avg total claims / meeting-model pair	Avg uncovered GT points / meeting-model pair	Avg GT points / meeting-model pair
gpt-41-mini	12.05	28.16	5.29	26.52
gpt-5-mini	16.05	37.27	3.20	26.30
gpt-51	18.00	40.45	1.21	26.35

gpt-41-mini says less and therefore leaves more GT uncovered. gpt-51 says the most and therefore leaves very little GT uncovered, but it also generates the largest number of unsupported or inaccurate claims. gpt-5-mini sits between these extremes, while gpt-41-mini retains the highest mean accuracy in the merged typed benchmark.

6.7 GT Validity Signals

The benchmark also exposes quality variation in the reference set. Within the merged typed benchmark, one meeting still contains a degenerate GT artifact with zero evaluable points (t_258810779752442698698651215247285387144), which propagates to six evaluator runs in the

merged setting. That case contributes no meaningful factual comparison and should be interpreted as a benchmark-quality warning rather than as model behavior. We retain it for transparency, but it underlines a core system point: GT quality assurance is part of the evaluation problem, not a preprocessing footnote. A second, softer signal is the whitehouse_press_briefings normalization difference noted earlier: those GT files retain structured topics and points but omit stored metadata counters, so the evaluator reconstructs counts by traversing the GT structure at runtime. That difference does not invalidate the benchmark, but it shows that reference normalization itself is an evaluable systems concern.

Inspectable artifacts make these warnings actionable rather than merely descriptive. An illustrative public case is whpb_20240110_press under gpt-41-mini, whose averaged accuracy is only 0.281. Inspection of the persisted accuracy_explanation payload shows a very specific pattern: roughly 31.0 claims per judge are flagged as unsupported_by_gt, compared with only 3.5 fabricated_facts and 0 direct contradictions, across an average of 48.0 extracted claims and 31.0 GT points. A scalar score alone would say only that this summary performed poorly. The persisted claim-level artifact shows something more diagnostic: a reference-side granularity mismatch dominated by unsupported specifics rather than a contradiction-heavy model failure. That kind of distinction is exactly why inspectable artifacts are treated as a first-class system goal in this paper. In practice, this diagnosis directs reviewer attention to GT granularity for this meeting type rather than to candidate-model behavior, which is the operational use case that file-backed persistence is designed to enable.

7. Discussion

7.1 What the System Operationalizes

The central value of the benchmark is not simply that it assigns scores. It turns transcript-GT pairs into reusable model-selection evidence. Because the GT universe, prompt pipeline, and reporting schema are held fixed, changes in results are interpretable as changes in candidate behavior rather than as changes in the evaluation setup.

This is also the clearest distinction between the present work and a standalone factuality scorer. A scorer can estimate whether one summary is faithful. The system described here additionally supports repeated comparisons under a fixed protocol, release decisions, aggregate error mining, typed benchmark analysis, and identification of benchmark bottlenecks such as weak or degenerate GT.

GT should therefore be understood as the reference rubric of the benchmark, not merely as another intermediate artifact. Candidate summaries are evaluated against GT rather than against raw transcripts directly. As a result, GT quality directly bounds evaluation quality: if GT omits salient content, correct candidate models can be penalized; if GT contains errors or broken structure, incorrect models can be rewarded.

The GT pipeline also clarifies where human review fits. In this system, human reviewers are not primarily judging candidate summaries directly; they are strengthening the reference rubric against which later candidate summaries will be scored. That distinction matters operationally. When human review

improves GT, it improves every downstream comparison on that transcript. When human review is skipped on a weak GT, scoring errors can propagate silently across all candidate models evaluated against it.

The same logic explains the audit stage. Audit is not another candidate-evaluation pass; it is a quality gate on the reference side of the pipeline. Its role is to detect missing topics, broken schemas or identifiers, shallow context, and other distortions in raw automated GT before those distortions propagate into downstream model scoring.

The typed benchmark is especially important here. If the system reported only merged averages, the White House slice would appear merely as a moderate overall accuracy decline. The typed partition instead reveals a distinct failure regime: all three models retain high completeness and coverage there, but all three become much worse on accuracy because they introduce more unsupported specifics. This is a systems lesson rather than a purely statistical one. Typed benchmarks do not only improve reporting; they change what kinds of failures the organization can see at all.

The comparison with private_data and city_council makes that diagnosis sharper. Those two slices achieve much higher accuracy not because they use a different candidate-generation prompt, but because their GT artifacts are finer-grained and better aligned with the fact structure encouraged by the application-facing summary prompt. In private_data, GT points retain explicit owners, dependencies, timing, blockers, and action items, and all benchmark GTs in that slice were human-reviewed after automated generation. In city_council, GT points and decisions retain motions, dates, recommendations, procedural outcomes, and amounts. Quantitatively, the average claim-to-GT ratio is close to 1:1 in private_data (1.05) and city_council (1.02), but much higher in whitehouse_press_briefings (1.41). The average number of inaccurate claims per meeting-model pair follows the same pattern: 11.3 in private_data, 9.1 in city_council, and 20.38 in whitehouse_press_briefings.

This contrast suggests that the White House accuracy problem is not primarily a prompt-quality failure. The same prompt family produces acceptable accuracy on private_data and city_council, where GT is sufficiently claim-complete to absorb transcript-faithful detail. In the private_data slice, that interpretation is strengthened by the fact that the entire GT subset was human-reviewed, which reduces the likelihood that its stronger accuracy is merely an artifact of weaker reference construction. The harder White House regime instead emerges from a pipeline mismatch: the application-summary prompt produces detailed, transcript-faithful public-affairs summaries; claim extraction then atomizes those details into many candidate claims; and the coarser White House GT cannot support many of those specifics even when they are grounded in the transcript. The typed benchmark therefore reveals a systems diagnosis rather than a single-model defect. The dominant problem is misalignment among transcript structure, GT granularity, and evaluation resolution.

7.2 What Pipeline Reuse Buys Across Domains

The broader systems contribution is cross-domain reuse. The AI-search instantiation and the meeting-summary instantiation differ in task semantics, but they share the same evaluation architecture:

- curated reference artifacts
- controlled candidate outputs
- automated scoring
- persisted explanations
- report-backed comparison for release decisions

This means the repository is not a single-task benchmark script. It is an evaluation substrate that can support multiple AI products by swapping task-specific references and metrics while preserving the orchestration logic. In practical terms, the reuse boundary is concrete. The search and meeting-summary instantiations share artifact persistence, batch orchestration, judge aggregation, report generation, significance reporting, and release-facing comparison logic. The components that change are the semantic layers: the reference schema, the generation prompts, and the task-specific evaluators. The meeting-summary case further shows that this substrate remains useful when both reference generation and score generation are themselves structured. In roadmap terms, the same architecture supports benchmarking, regression detection, and failure-case analysis without rebuilding the full control plane for each product domain.

Component layer	AI-search instantiation	Meeting-summary instantiation	New-task replacement needed
Artifact persistence	File-backed search artifacts	File-backed transcript/GT/candidate/evaluation artifacts	No
Batch orchestration	Automated benchmark execution	Automated benchmark execution	No
Judge aggregation	Aggregated retrieval results and reports	Aggregated factual-metric results and reports	No
Significance reporting	Search-comparison reporting	Meeting-level sign-test reporting	No
Release-facing report generation	Search model/pipeline comparison outputs	Summary model comparison outputs	No
GT schema	Query-document relevance references	Typed meeting artifacts	Yes
GT generation prompts	Relevance/QA-style synthesis prompts	Multi-stage meeting-structure extraction prompts	Yes
Metric evaluators	Retrieval metrics	Accuracy, completeness, and coverage evaluators	Yes

To make the reuse boundary concrete: migrating from the AI-search instantiation to the meeting-summary instantiation required replacing three semantic layers while keeping five orchestration layers intact. The changed components are the GT schema (from query-document relevance labels to typed meeting artifacts), the GT generation prompts (from relevance annotation to multi-stage meeting-structure extraction), and the metric evaluators (from retrieval-oriented comparison to accuracy, coverage, and completeness). The reused components include artifact persistence, batch execution, judge aggregation, significance testing, and release-facing comparison logic. This boundary is the operational meaning of cross-domain pipeline reuse in this paper.

7.3 Implications for Model Selection

The benchmark supports a policy-driven interpretation. In the merged typed benchmark, gpt-41-mini has the highest mean accuracy, but the differences are not statistically significant. If the product objective is retention-first, gpt-51 is preferable because its completeness and coverage advantages are both larger and statistically supported. If a deployment wants a single scalar ranking, the mean-of-metrics score favors gpt-51, but that choice should be made explicitly because it accepts lower accuracy in exchange for better retention.

This result is also useful as an engineering decision rule. “Not statistically significant” does not mean “irrelevant”; it means the benchmark does not support a stable accuracy winner under the present paired design. A disciplined release policy can therefore treat accuracy as a quality floor and secondary monitoring metric, while reserving primary selection logic for metrics whose advantages are both large and significance-backed. In other words, the system helps separate what is directionally interesting from what is robust enough to gate releases.

The focused gpt-5.4 versus gpt-4.1 comparison sharpens that deployment rule for workloads dominated by private_data and city_council. Because that follow-up was executed under the same pipeline, judges, and metric definitions, it should be read as operational reuse of the same evaluation system under a different deployment question rather than as an apples-to-oranges side benchmark. In that operational slice, gpt-5.4 improves completeness and coverage by large, significance-backed margins while also moving accuracy in the favorable direction under the same evaluation protocol. For teams choosing between those two models on enterprise-style and civic-proceedings data, the benchmark therefore supports a cleaner recommendation than the broader mixed benchmark does: gpt-5.4 is the preferable default, because it retains substantially more actionable content without paying a measured accuracy penalty.

7.4 Relationship to Claim-Based Evaluation Work

Relative to claim-based meeting-summary work, this paper shares the same factual core: summaries are judged against grounded factual units rather than only style or fluency. The difference is scope and structure. Most claim-based papers contribute a benchmark, a factuality metric, or an evaluator study. This paper contributes an implemented quality system around that core: structured GT construction, structured score generation, candidate generation, judge disagreement analysis, typed benchmark breakdowns, persisted reports, and release-facing comparison under a reusable cross-domain architecture. The same distinction also explains the architectural comparison in Table 0. General-

purpose frameworks provide useful evaluation components, but this system treats reference construction, typed benchmarking, and report persistence as part of the evaluation product rather than as surrounding glue code.

The DeepEval baseline should be read under that same boundary. It is valuable here as a ranking-consistency check under a different evaluation philosophy, not as a controlled head-to-head systems comparison establishing absolute superiority. The empirical point is narrower and more defensible: holistic reference scoring preserves the broad retention ordering, but it does not expose the same claim-grounded failure structure or release-facing diagnostics that this system is designed to surface.

7.5 Future Work and Improvement Plan

This section focuses on forward-looking system improvements rather than restating current limitations. The White House anomaly points to a concrete improvement path. The first priority is typed GT densification. For briefing-style data, GT should preserve more of the entities, dates, quantities, organizations, and policy subclauses that are routinely expressed in transcript-faithful summaries. This is not only an annotation expansion step; it is a benchmark-quality intervention that reduces the gap between what the product-style prompt naturally summarizes and what the evaluator can legitimately score.

The second priority is typed prompting rather than one-prompt-for-all evaluation. The current benchmark intentionally fixes a single application-facing prompt family across all dataset types in order to isolate model effects. That choice is appropriate for comparison, but the typed results suggest that future production-oriented configurations should use prompt specializations by data regime. private_data and city_council benefit from prompts that foreground actions, owners, dates, and decisions. whitehouse_press_briefings likely needs a more conservative public-affairs variant that summarizes stance and topic structure without overcommitting to every policy qualifier or secondary numeric detail.

The third priority is to make the evaluator more explicit about transcript-true but GT-omitted details. At present, many such claims are absorbed into unsupported_by_gt or fabricated_facts, which is useful for conservative GT-based benchmarking but less useful for separating candidate error from GT incompleteness. A future score schema should therefore distinguish at least three conditions: genuinely inaccurate claims, GT-supported claims, and transcript-supported-but-GT-omitted claims. This would preserve strict benchmarking while improving diagnosis.

The fourth priority is deeper structured candidate generation. One reason this repository is reusable across domains is that both GT and score generation are structured. The same principle can be extended to candidates by introducing an intermediate structured summary ledger over entities, actions, decisions, dates, and quantities before the final natural-language rendering step. That intermediate representation would likely stabilize both summary generation and downstream claim extraction, especially in fact-dense slices such as whitehouse_press_briefings.

Finally, future work should validate improvements through typed ablations rather than only merged reruns. A practical next experiment is a 2 x 2 White House study: current GT vs denser GT, and current prompt vs a briefing-specialized prompt. Such a design would make it possible to attribute gains to benchmark improvement, prompt adaptation, or both. More broadly, the system should continue to use typed benchmarks as the control surface for quality iteration, because the central lesson of this paper is that merged averages alone hide diagnostically important failure regimes.

8. Limitations

This section isolates the current benchmark's objective constraints. Unlike Section 7.5, it does not propose improvement paths; it records what the present study cannot yet claim.

8.1 GT Quality Bounds the Evaluation

If GT omits true details or compresses them inconsistently, some unsupported_by_gt findings may partly reflect reference limitations rather than pure candidate error. In practical terms, a GT that misses content can penalize a correct model, while a GT with errors can reward the wrong model. The current GT path relies primarily on multi-stage model agreement and optional automated audit rather than mandatory human review, so the benchmark inherits the strengths and weaknesses of automated structured GT construction. One important exception is the private_data slice, for which all 22 GT artifacts were human-reviewed after automated generation. The degenerate GT case in the benchmark nevertheless makes this threat concrete for the benchmark as a whole.

8.2 LLM Judges Still Add Variance

Two judges improve robustness, but accuracy judgments remain somewhat unstable. Accuracy-sensitive release decisions should therefore consider both mean score and judge spread.

8.3 Internal Benchmark Slice

The 114-meeting merged typed benchmark is substantially stronger than a toy sample, but it still reflects the repository's current data mix. The results should be read as evidence about this system and these three meeting-data categories, not as universal conclusions about all meeting-summary tasks. The private_data subset contains only 22 meetings due to data-access constraints. At this sample size, subset-level significance claims have limited statistical power; the reported patterns for this slice should be treated as descriptive evidence rather than as significance-backed findings in isolation. Expanding that slice is an explicit target for future benchmark growth. The whitehouse_press_briefings slice behaves like a distinct sub-problem: the summaries become much less accurate there without a corresponding collapse in coverage or completeness. That makes the slice diagnostically valuable, but it also means the benchmark mixes multiple meeting-summary regimes rather than one homogeneous task.

8.4 Fixed Prompt Family

This study intentionally holds the summary prompt pipeline fixed at the current implementation (standard_v1 + format_v1). The results therefore compare models under one prompt family rather than under each model's best possible prompt tuning.

8.5 Conservative Significance Test

The sign test fits the paired meeting-level comparison setting, but it ignores score magnitude. It therefore complements rather than replaces the descriptive mean differences reported earlier. One comparison deserves special attention: gpt-41-mini versus gpt-51 on accuracy yields a Holm-adjusted p-value of 0.053, which falls just outside the 0.05 threshold after correction. Under the corresponding unadjusted test (exact two-sided sign test $p = 0.0178$ before family-wise correction), this comparison would be significant, but Holm correction appropriately penalizes the three-comparison family. We therefore treat this result as directional evidence rather than a confirmed finding: with a larger benchmark slice, the observed accuracy advantage of gpt-41-mini over gpt-51 may become significant.

8.6 Limited Empirical Baseline Comparison with General-Purpose Evaluation Frameworks

This paper now includes a typed empirical comparison against DeepEval across all three dataset slices, which is sufficient to show that holistic reference scoring and structured claim-grounded scoring preserve the same retention ordering while diverging substantially on accuracy, especially in the White House slice. However, the comparison is still intentionally lightweight rather than fully standardized: it does not align evaluator prompts, metric semantics, or aggregation rules across systems. In particular, the DeepEval baseline uses gpt-4.1-mini as its evaluator model, whereas our system uses gpt-5 and Claude Sonnet 4 as judges, creating an evaluator-model confound that prevents score-level interpretation. Because the evaluator model differs, score-level comparison is not valid; the baseline is therefore used only to verify ranking consistency rather than to calibrate absolute scores. A broader head-to-head against systems such as RAGAS, TruLens, and DeepEval with matched semantics remains future work.

8.7 No Benchmark-Wide Cost Table Yet

The repository now supports two telemetry views: a measured view recoverable from retained historical offline-evaluation logs, and an instrumentation path through log_config.metrics for future normalized collection. This paper reports the measured offline-evaluation telemetry that can be recovered from retained logs, but those artifacts do not extend cleanly to GT construction and candidate generation across the full benchmark. The system is therefore beyond a purely estimated cost story, but it still does not yet provide a complete measured end-to-end latency/cost ledger for the entire five-stage pipeline.

The same boundary applies to the implemented online path. Although the deployed architecture includes feedback ingestion, evaluation triggering, result persistence, and downstream metrics emission, the present paper does not yet report usage volume, latency, retry/failure behavior, or benchmark-refresh effects for that component. The online path should therefore be read as implemented system scope, but not yet as a quantitatively characterized one.

8.8 Partial Reproducibility Constraint

The private_data subset (22 meetings) cannot be released in raw form because of enterprise confidentiality constraints. Raw transcripts, GT artifacts, and candidate summaries for that slice are therefore excluded from the packaged distribution. The aggregate benchmark CSVs retain the private_data rows, so the reported statistics are complete, but independent replication of the private_data slice requires access to comparable internal meeting data. This constraint applies even though the private_data GTs are stronger than the default benchmark path in one respect: all 22 GT artifacts in that slice were human-reviewed after automated construction. Reviewers who need to verify the private_data results may contact the authors for summary-level artifact inspection under NDA and subject to internal confidentiality review.

8.9 Cross-Domain Reuse Demonstrated Across Two Papers, Not Within a Single Controlled Migration

The cross-domain reuse claim in this paper is supported by two concrete instantiations: the AI-search instantiation documented in the companion paper (Zhong et al., 2025) and the meeting-summary instantiation documented here. The reuse table in Section 7.2 identifies which components were replaced and which were inherited. However, this manuscript does not include a controlled migration experiment to a third independent task domain within the same paper. Cross-domain reusability is therefore supported by design evidence and two domain instantiations rather than by an independent replication study.

9. Conclusion

We presented the Dataset Pipeline repository as a reusable cross-domain evaluation system whose current benchmarked instantiation targets AI meeting summaries and whose earlier instantiation supported AI search. On a merged typed benchmark of 114 meetings, 340 completed meeting-model pairs, and 680 evaluator runs, the system reveals a stable trade-off: gpt-41-mini is strongest on mean accuracy, whereas gpt-51 is strongest on completeness and coverage. Paired sign tests show that the accuracy differences are not statistically significant after correction, while the retention advantages of gpt-51 are significant. A focused operational-slice comparison further shows that this frontier is workload-dependent: on private_data and city_council, gpt-5.4 is descriptively stronger than gpt-4.1 on all three metrics, with statistically robust gains on the retention-oriented metrics.

Three broader conclusions follow. First, evaluation-pipeline reuse is practical across heterogeneous AI tasks when orchestration is separated from task-specific reference schemas and scoring layers. Second, the meeting-summary instantiation is strongest when both GT generation and score

generation are structured, because that structure supports claim alignment, typed analysis, significance testing, and release-facing reporting. Third, typed benchmarks function as diagnostic instruments: without dataset-type partitioning, the White House accuracy failure regime would have been obscured by merged averages, appearing only as a moderate overall decline rather than as a structurally distinct pipeline-mismatch problem. The DeepEval baseline reinforces that same point from the opposite direction: holistic scoring preserves ranking consistency, but without structured claim-grounded evaluation it does not surface the unsupported-specifics and GT-granularity mismatch that define the White House failure regime. More generally, the results show how pipeline-oriented evaluation can move from AI search to meeting summarization without changing its operational core, while leaving quantitative characterization of the implemented online loop as a concrete next step for production-feedback integration.

10. Reproducibility Pointers

The full repository implementation underlying this paper is developed in an internal Cisco repository, but the artifact package for this paper, including the benchmark CSVs, derived reports, and main driver programs referenced below, is mirrored in the publicly accessible paper repository Paper. Public repository URL: https://github.com/Philipzhong1980/cross-domains-evaluation-framework. This repository is publicly accessible without Cisco-internal credentials. To respect privacy constraints, the packaged dataset/ subtree includes the city_council and whitehouse_press_briefings raw artifacts with their original repository paths preserved, but it intentionally excludes the raw private_data meeting contents. The combined benchmark CSVs are retained unchanged, so their aggregate statistics still reflect the full benchmark, including the private slice. Unless otherwise noted, manuscript tables are computed from meeting_model detail rows under the aggregation rule stated in Section 5.4 rather than from the convenience summary rows embedded at the top of some packaged CSVs. The package is therefore artifact-complete and reproducible for the public-sector slices, subject to external model/API access, but only partially reproducible for the full merged typed benchmark; independent verification of the private_data slice requires equivalent internal data access.

Teams seeking to instantiate this pipeline for a new generative-AI evaluation task should treat the published implementation files as entry points rather than as a monolith requiring fork-and-rewrite. The task-specific layers that require replacement are: (1) the GT schema and GT generation prompts, (2) the candidate generation configuration, and (3) the metric evaluators. The orchestration, persistence, aggregation, significance testing, and release-reporting layers can be reused unchanged. In practical terms, a new task instantiation primarily means authoring task-specific schemas and prompts rather than rebuilding the evaluation infrastructure.

The core artifacts for submission are:

- Latest manuscript: meeting-summary-system-paper_v24.md, meeting-summary-system-paper_v24.html, meeting-summary-system-paper_v24.pdf
- Public artifact repository for this paper: Paper
- Core implementation paths mirrored for public release: meetingsummary_ground_truth.py, meetingnotes_candidate.py, offline_evaluation_service.py

- Main merged benchmark report and significance analysis: meeting_notes_model_comparison_combined_20260417.csv, meeting_notes_model_comparison_combined_significance_20260417.csv
- Focused gpt-5.4 vs gpt-4.1 follow-up bundle: README.md, meeting_notes_model_comparison_combined_20260422_134801.csv, meeting_notes_model_comparison_gpt54_vs_gpt41_significance_20260422.csv, meeting_notes_model_comparison_gpt54_vs_gpt41_analysis_20260422.md, run_report_20260422.md
- Baseline and supporting analysis code: batch_meeting_notes_model_comparison.py, deepeval_typed_empirical_comparison.py
- Typed DeepEval baseline report: deepeval_typed_empirical_comparison_all_20260421.csv
- GT agreement statistics: gt_agreement_stats_all_20260421.csv
- Single-vs-two-judge ablation: single_vs_two_judge_ablation_all_20260421.csv
- Measured telemetry from retained logs: measured_stage_level_telemetry_from_logs_20260421_131115.csv
- Measured telemetry export code: export_measured_telemetry_from_logs.py
- Architecture figures: dual_loop_quality_framework.svg, repository_pipeline_implementation.svg
- Packaged dataset subtree: dataset

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Entries 1-10 are scholarly references. Entries 11-13 are software/documentation sources cited for the architectural capability comparison in Table 0.

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